

# 1.1 Chemistry of Life — Macromolecules

Unit 1: Cellular Functions

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## Overview

All living organisms are built from the same core chemical elements: carbon (C), hydrogen (H), oxygen (O), nitrogen (N), phosphorus (P), and sulfur (S). These elements bond to form four major classes of biological macromolecules — large molecules essential for life. Understanding these molecules is the foundation of cell biology.

## The Four Macromolecules

### ■ Key Concept

The four macromolecules of life are: Carbohydrates, Lipids, Proteins, and Nucleic Acids. Each is built from repeating subunits called monomers that link together to form polymers.

## Key Vocabulary

| Term                         | Definition                                                                                                                       |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Monomer</b>               | A single repeating subunit that links with others to form a polymer (e.g., glucose is a monomer of starch).                      |
| <b>Polymer</b>               | A large molecule made of many monomers bonded together (e.g., cellulose is a polymer of glucose).                                |
| <b>Carbohydrate</b>          | Molecules made of C, H, O in a roughly 1:2:1 ratio. Functions: energy source and structural support. Monomer = monosaccharide.   |
| <b>Lipid</b>                 | Non-polar molecules including fats, phospholipids, and steroids. Functions: energy storage, insulation, cell membrane structure. |
| <b>Protein</b>               | Polymers of amino acids. Enormously diverse in function: enzymes, structural proteins, transport, immune defense, hormones.      |
| <b>Nucleic Acid</b>          | DNA and RNA — polymers of nucleotides that store and transmit genetic information and direct protein synthesis.                  |
| <b>Dehydration synthesis</b> | The reaction that joins monomers into polymers by removing a water molecule.                                                     |
| <b>Hydrolysis</b>            | The reverse reaction — a polymer is broken apart by adding water.                                                                |

## Carbohydrates in Detail

### ■ Tip

Positive test for reducing sugars (monosaccharides and some disaccharides): Benedict's solution turns brick-red/orange when heated. Positive test for starch: iodine turns blue-black.

Monosaccharides (glucose, fructose, galactose) are the simplest sugars. Disaccharides (sucrose = glucose + fructose; lactose = glucose + galactose) are two monosaccharides joined. Polysaccharides are long chains: starch and glycogen store energy; cellulose (in plants) and chitin (in fungi/insects) provide structural strength.

## Lipids in Detail

### ■ Tip

Positive test for lipids: Sudan IV stain turns red/pink.

Triglycerides (fats and oils) consist of one glycerol + three fatty acid chains. Saturated fatty acids have no double bonds (solid at room temp); unsaturated have one or more double bonds (liquid at room temp). Phospholipids have one fatty acid replaced by a phosphate group — making one end hydrophilic and one hydrophobic, critical for membrane structure. Steroids (e.g., cholesterol) have a four-ring carbon skeleton.

## Proteins in Detail

### ■ Tip

Positive test for proteins: Biuret reagent turns purple/violet in the presence of peptide bonds.

Proteins are polymers of amino acids linked by peptide bonds. The sequence of amino acids (primary structure) determines how the protein folds into its unique 3D shape — and therefore its function. A change in even one amino acid can destroy a protein's function (e.g., sickle cell disease). Enzymes are proteins that act as biological catalysts — they speed up reactions without being consumed.

## Key Diagrams to Know

- Glucose ring structure ( $\alpha$ -glucose vs.  $\beta$ -glucose)
- Phospholipid structure: hydrophilic head (glycerol + phosphate) and two hydrophobic tails (fatty acid chains)
- Amino acid general structure: central carbon bonded to amino group ( $\text{-NH}_2$ ), carboxyl group ( $\text{-COOH}$ ), hydrogen, and variable R-group
- Nucleotide structure: phosphate group + 5-carbon sugar (deoxyribose in DNA, ribose in RNA) + nitrogenous base

## Summary Comparison Table

| Term          | Definition                                                             |
|---------------|------------------------------------------------------------------------|
| Macromolecule | Monomer   Element   Test   Function                                    |
| Carbohydrate  | Monosaccharide   C, H, O   Benedict's / Iodine   Energy; structure     |
| Lipid         | Fatty acid + glycerol   C, H, O   Sudan IV   Energy storage; membranes |
| Protein       | Amino acid   C, H, O, N, S   Biuret   Enzymes; structure; transport    |
| Nucleic Acid  | Nucleotide   C, H, O, N, P   —   Genetic info; protein synthesis       |